A proposed 392-variable quartic refinement

Contact-unit normalization, an outer scalar constant,
and coefficient balancing without extra variables

Prepared with ChatGPT for Cameron Freer

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1 Result, provenance, and scope

The selected proposal has **392 total rational variables**, including the unchanged input X, and therefore 391 existential witnesses. Its 360 quadratic residuals give a polynomial of degree exactly four with 26,301 nonzero monomials. Every coefficient is written explicitly in decimal. Every coordinate occurs; the coefficient of X^4 is one, and the primitive integer content is one.

After a certified diagonal change of twelve existing coordinates, its largest absolute coefficient is exactly

$$1 + 405323966463344643^{1860}.$$

This has **905 decimal digits and 3004 bits**. No quantified constant-chain coordinate is needed for this form. The fixed coefficient in the underlying arithmetic source is $S_0 = 3^{2790}$.

Form	Variables	Residuals	Monomials	Max. digits
Balanced direct decimal	**392**	360	26,301	**905**
Unbalanced direct decimal	392	360	26,301	2,679
Two-coordinate constant lift	394	362	26,310	888
Three-coordinate constant lift	395	363	26,313	381
Four-coordinate constant lift	396	364	26,316	178

Compared with the preceding proposal, the core count drops from 396 to 392. At the preceding 978-digit coefficient cap, the count drops from 398 to 392, and the maximum coefficient also becomes smaller. The monomial count rises by 76. Against the pinned public Version 10 counts, the changes are $523 \to 392$ and $540 \to 392$ at a smaller maximum coefficient size than its decimal table.

The intended characterization is **conditional on the inherited global arithmetic inputs**:

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists (y_1, \ldots, y_{391}) \in \mathbb{Q}^{391}, \quad P_{392}(X, y) = 0.$$

This proposal supplies local replacement arguments and exact finite algebraic certificates. It does not independently establish the inherited curve/group assertions, geometric and quaternionic inputs, reduction, modularity and local compatibility, global height theorem, simultaneous prime/progression results, Lucas theorem, or pole realization. The upstream qualifications about inspected source versions remain in force. Internal review by different AI agents is not human peer review or a completed Lean proof. No optimality or historical priority is asserted.

1.1 The three improvements

Multiplying each semantic detector test by its contact-unit guard changes the weight from e^{2m-1}/δ^3 to e^{2m-1}/δ. This permits direct computation of the order-one weight and removes the separate coordinate for e: one saving in each of three detector copies.

The scalar block uses the actual input polynomial to show that an odd contact valuation is already positive. Consequently its constant one can be placed outside the norm tree. This removes one coordinate while preserving the scalar order twenty-eight and real positivity.

Finally, a sharper use of the existing numerical height constant permits $S_0 = 3^{2790}$. Writing $S_0 = c^6$ allows a simple diagonal change with factors c, c^2, c^4. It distributes the large anchor coefficient through existing equations and removes the need for constant-chain variables at the preceding coefficient cap.

The complete selected pipeline measures

$$403 \text{ coordinates}/371 \text{ rows} \longrightarrow 392/360.$$

The independent replay checks 342 source gate identities and 333 compiler operations. There are zero additional eliminations after the generic compiler; earlier eliminations are not counted a second time. The optional constant lifts are unique triangular rational extensions, with exponent lists $(930, 930)$, $(399, 399, 396)$, and $(186, 186, 186, 186)$.

1.2 Pinned sources and publication status

The public baseline is `elliotglazer/diophantine-z-q`, commit `036cebe4f1a4aa7ca89820a9b48f3629b4c364d6`. The selected builder has SHA256 `78b54bcde24ff8a91091e102a4734c49878699f071d9230e5ce7a70ffd2114d0`. The repository credits, including Gabriel Conant with ChatGPT and the collective agentic improvements, are retained. The draft PR package contains the prior local proofs and a dependency/consumer inventory. The current four proof sections identify their specific inherited inputs.

This is an additive research proposal. It does not change the current upstream manuscript, claim an official Version 11, or rebind the frozen Version 10 audit acceptance. Manuscript incorporation remains pending. No public PR, push, release, or external message was made in preparing the package. The included handoff is intended for a separate coding-agent session, with an apply-checkable patch and a draft PR description.

2 Rescale the local tests and remove the intermediate weight

The starting source is the frozen sixth-round construction, with builder SHA256

`3f63cb782d52857446cf39f6e545aeb74f380d1347826eafa888cbb487738d51`.

The selected integrated source has SHA256

`78b54bcde24ff8a91091e102a4734c49878699f071d9230e5ce7a70ffd2114d0`.

The change in this section removes one surviving rational coordinate from each of the three detectors. It leaves the orders $1, 12, 45$, all sixteen underlying field tests, the logarithms, height scales, contact coordinate, and pole predicate unchanged. The global arithmetic qualifications are inherited; the construction is not asserted to prove those inputs independently.

2.1 The normalization freedom in the actual local lemma

Retain

$$t = A\lambda, \qquad \delta = 98t - 1, \qquad F(t) = t(t - 1/49)(t - 1/50)(t - 1/2450)(t - 32/1225),$$

and retain the semantic quantity

$$R = \frac{F(t)}{A^5\delta^6}.$$

The three scales are still $1, C_0^2, (C_0 + C_1)^2$. On the source domain they are positive rational squares and at least one. The positive chart for λ gives $t \geq 32/1225$, hence $\delta \geq 39/25 > 0$.

At every prime consumed by the retained five-contact height argument, δ is a unit. The values at the five contacts are

$$-1, \quad 1, \quad 24/25, \quad -24/25, \quad 39/25.$$

Their prime support is already contained in $\mathcal{S} = \{2, 3, 5, 7, 13, 17\}$. In particular 13 remains exceptional; we do not claim that the guard is a unit there. A pole of t is not a consumed contact outside this set: its image under $\phi(t) = -49 + 1/t$ reduces to -49, distinct from the five height contacts.

Let D_j be an existing test of order m_j. In the arbitrary-order one-place lemma, replace its semantic test by

$$D_j' = \delta D_j.$$

No coordinate for this product is required. After the original normalization, the actual factor is

$$1 + R\left(\frac{D_j/\delta^{m_j}}{R^{m_j}}\right)^2 = 1 + \frac{e^{2m_j-1}}{\delta}D_j^2, \qquad e = \frac{(A\delta)^5}{F(t)}.$$

Thus use the new weights

$$q_m = \frac{e^{2m-1}}{\delta}$$

in place of e^{2m-1}/δ^3. The rational function e is useful for explaining the identity, but no longer needs its own coordinate.

This is an application of the actual proof of the one-place lemma, `manuscripts/quartic_v10.tex`, lines 11031–11061, in the pinned upstream source. Its field inputs are arbitrary elements of K; it requires no integrality assumption on D_j/δ^m before the conclusion. At a consumed prime the unit property gives

$$v_{\mathfrak{p}}(D_j/\delta^m) = v_{\mathfrak{p}}(D_j),$$

so the conclusion is exactly the original order requirement for D_j. All sixteen orders still hold at one common place. No new formal-group claim, prime-selection theorem, or logarithm precision assertion is being substituted for the retained one-place argument.

2.2 Compute the weights without storing e

Keep the existing $U = A\delta$ and its fourth power. Define directly

$$q_1 F = AU^4, \qquad E = q_1\delta^2.$$

These give

$$q_1 = \frac{A^5\delta^4}{F} = \frac{e}{\delta}, \qquad E = \frac{A^5\delta^6}{F} = R^{-1}, \qquad q_m = E^{m-1}q_1^m.$$

The value $\delta^2 = 9604t^2 - 196t + 1$ is affine in the already stored t^2 and t. The implementation requests its square through the common-subexpression routine, which reuses the existing quadratic part. It adds no new coordinate for this square.

The order-one product evaluation already requires q_1^2, q_1^3, q_1^4. Using its actual stored cube, compute the two other weights in seven gates:

$$\begin{aligned} a &= E^2, & b &= aE = E^3, & c &= bq_1^3, & d &= c^2, \\ q_{12} &= d^2/E, & h &= q_{12}^2, & q_{45} &= h^2/q_1^3. \end{aligned}$$

The exponents in (E, q_1) are respectively

$$(2,0), (3,0), (3,3), (6,6), (11,12), (22,24), (44,45).$$

Equivalently, their exponents in the explanatory pair (e, δ) are

$$(2,2), (3,3), (6,0), (12,0), (23,-1), (46,-2), (89,-1).$$

The last two outputs are therefore exactly the required new weights. The same grouping of the field factors is retained: one mixed $(1, 12)$ block, two four-test order-one blocks, and two three-test order 45 blocks. All shared symmetric coefficients and the mixed-block identity remain unchanged.

The old implementation separately stored e, δ^3, and the quotient $q_1 = e/\delta^3$. The new one defines q_1 directly and never uses a stored e or δ^3. The latter three coordinates were already removed by the previous final elimination stage, so their raw removal does not itself constitute a final saving. The net measured saving is the other three quotient coordinates, one per detector.

2.3 Nonzero domains and quadratic degree

On the positive chart, all factors of F are nonnegative. The relation $q_1 F = AU^4$, whose right side is strictly positive, excludes $F = 0$. Consequently F, q_1, E are positive. Every intermediate in the displayed chain is positive, and its two quotient denominators E and q_1^3 are nonzero. Each auxiliary coordinate is uniquely determined by the preceding coordinates. Eliminating this chain recovers the exact rational weight formulas; conversely those formulas uniquely extend every point of the established nonzero domain.

Every displayed gate equation has degree at most two in stored coordinates. For example $q_1 F = AU^4$ uses the already stored fourth power, $Eq_{12} = d^2$, and $q_1^3 q_{45} = h^2$ uses the already stored cube. The source checker verifies the actual square/cube aliases reused by the order-one evaluation, rather than merely counting them as free.

Each factor $1 + q_m D_j^2$ is at least one at both real embeddings. If $W = W_0 + W_1\sqrt{2}$ is their product, then $W_0 \geq 1$ and $N(W) > 0$. The stored norm quotient and pole input remain

$$n = \frac{N(W)}{2W_0}, \qquad B = \frac{n^2}{E} = \frac{R}{\mathrm{Tr}(W^{-1})^2}.$$

Their denominators are nonzero. The pole input uses exactly the old R, not a modified rational function differing by an unchecked factor. The normalization changes the existential detector witnesses; it is justified by soundness and completeness transfer, not by a claimed bijection between every old and new pole witness tuple.

2.4 Completeness at contacts, other primes and infinity

The preparation and order of choices remain those of the source. First fix the input, scalar preparation, four root points and rational square scales. The source real-progression argument supplies bounds on the unscaled D_j independent of future progression moduli. Its finite neighborhood preparation supplies common lower valuation bounds on D_j above $2, 3, 17$. The exact relevant passage is the pinned `quartic_v10.tex`, lines 11220–11241. The same prime selection and primary progression $\Pi^{44} \mid h$ are then used. Alternate points still require only formal depth one.

At a selected contact prime, R has valuation one and δ is a unit. Multiplying the old factor's nonconstant term by δ^2 therefore does not change its valuation. At the successful place every factor is one modulo the prime. At either place every inverse factor has residue zero or one, because its nonconstant term has odd valuation. Hence the reciprocal trace has residue one or two and is a unit. The new B has the same positive odd contact valuations as required.

The exact product formula for R is unchanged:

$$R = \frac{\prod_{\theta\in\Theta}(u - (\theta/A)v)}{v^5\delta^6}, \qquad v = D_0^2 w^2.$$

The denominator is a rational square and the numerator consists of the selected contact primes, with the existing incidence convention for repeated forms. At every other prime $v_p(R)$ is even, so $v_p(B)$ is even regardless of the trace valuation. Thus the reduced denominator of B is a square. In particular a new zero of δ away from contacts is harmless: its contribution to R has even order. The fixed Gaussian classes still make R, and hence B, square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$.

For fixed A and $\lambda \to +\infty$,

$$e \longrightarrow 98^5 A^5 > 0, \qquad \delta \sim 98A\lambda, \qquad q_m = O(\lambda^{-1}).$$

The uniform bounds on the unscaled tests therefore give

$$W = 1 + O(\lambda^{-1}), \qquad \mathrm{Tr}(W^{-1}) = 2 + O(\lambda^{-1}), \qquad B \sim \frac{1}{4 \cdot 98^6 A^6 \lambda}.$$

This is slower decay than before, but it still gives $0 < B < 17$ for sufficiently large prescribed λ. The actual preparation-zero pole converse requires this bound; no stronger real assertion is used.

At a fixed small prime $p = 2, 3, 17$, put

$$\nu = v_p(v), \qquad \eta = v_p(A), \qquad \alpha = v_p(98).$$

The scale and α are fixed before choosing large ν. Since u is a selected prime, $v_p(u) = 0$. For all sufficiently large ν, dominance of the leading terms gives

$$v_p(\delta) = \eta + \alpha - \nu, \quad v_p(e) = 5(\eta + \alpha), \quad v_p(q_m) = \nu + (10m - 6)(\eta + \alpha).$$

If $v(D_j) \geq -c_p$ on the fixed neighborhood, then

$$v(q_m D_j^2) \geq \nu + (10m - 6)(\eta + \alpha) - 2c_p \longrightarrow +\infty.$$

Thus the trace is as close to two as necessary, including at the ramified prime two. Since R is unchanged,

$$v_p(B) = \nu - 6(\eta + \alpha) - 2v_p(2).$$

Prescribed high divisibility of w gives the same required bounds $v_2(B) \geq 3$, $v_3(B) \geq 1$, $v_{17}(B) \geq 2$. The unchanged preparation-zero bridge therefore supplies its four pole witnesses. This completes the changed local and real completeness checks.

2.5 Actual counts and verification boundary

The isolated replacement, with all other sixth-round formulas and $S_0 = 3^{3072}$ retained, compiles as

$$\text{raw } 404/372 \longrightarrow \text{generic } 393/361 \longrightarrow \boxed{\text{final } 393/361}.$$

The pairs count rational coordinates and residuals. No further elimination is needed after the generic compiler. In particular the unit-scale fourth-power elimination is now carried out there, and no old final-elimination saving is counted twice.

With the separately proved scalar outer-constant change and selected anchor $S_0 = 3^{2790}$, the integrated source compiles as

$$403/371 \longrightarrow \boxed{\text{392 coordinates and 360 residuals}}.$$

Again there are zero further eliminations. The fixed-parameter quartic has 26,301 ordinary-coordinate monomials, degree four and primitive integer content one. Its coefficient balancing and anchor margins are separate transformations with their own exact certificates and proofs.

The independent checker `check_detector_round7.py` imports neither producer nor compiler. On the frozen integrated source it passes 384 source assertions, seven Laurent exponent identities, three semantic normalization identities, and two expanded mixed-block identities. It reconstructs the retained G_{12}/G_2 logarithms, all test aliases, all shared coefficients and product evaluations, the direct q_1 equation, $E = R^{-1}$, every weight gate, actual reused powers, norm and pole consumers, the contact and positivity data, and the lifted-degree bound. The global arithmetic implications remain qualified as in the source; these finite checks do not prove those inherited results.

2.6 Rejected searches and the useful obstruction

A bounded exact exponent search for six-node replacements of the old weight chain found no candidate in its stated restricted class. This is not a general minimality result. There is nevertheless a useful simple obstruction in the old monomial-gate model. With the existing monomials $e, \delta, \delta^2, \delta^3, E, q_1^2, q_1^3, q_1^4$, where the old $q_1 = e/\delta^3$, the functional $L(a, b) = 13a + 3b$ on exponent pairs has all input values in $[0, 16]$. A quadratic quotient operation $A + B - C$ can at most double the width of the current interval. After six operations its maximum cannot exceed 1024, whereas $L(q_{41}) = 1044$. Thus a six-node monomial chain cannot even produce that high-order weight from these inputs. The search also considered lower orders and shared outputs, without claiming an exhaustive result for arbitrary rational circuits.

The selected improvement changes the normalization, which changes these input monomials and allows removal of the stored e; it does not claim to beat that restricted lower bound. Scaling instead by $\delta^2 D$ would make the weights grow at real infinity and generally defeats the preparation-zero bound $B < 17$ without a further modification. That variant is not used. The selected δD scaling retains decay and has been checked against the actual consuming inequalities above.

3 Moving the scalar constant outside the norm tree

This replacement starts from the frozen sixth-round construction with builder SHA256

`3f63cb782d52857446cf39f6e545aeb74f380d1347826eafa888cbb487738d51`.

It removes one rational coordinate after the complete compiler while preserving scalar order twenty-eight. The point charts, nonzero shifted-point guard, scalar proxy, logarithms, detector tests, physical anchor, and scalar pole equations are unchanged. The new scalar witnesses are justified by a soundness and completeness transfer; a bijection between the old and new scalar witness tuples is neither needed nor asserted. The inherited global arithmetic remains conditional.

3.1 The replacement and its quadratic implementation

Retain the actual input polynomial and contact product

$$\beta = 2401 + 2754(2X+1)^2 = 11016X^2 + 11016X + 5155, \qquad r = \beta(\beta-1)(32\beta+343).$$

For every rational input, $\beta \geq 2401$ and $r > 0$. Retain the scalar points $H = [h]P$ and $Q = S - P = [a-1]P$, where $h \neq 0$ and $a - 1 \neq 0$, and write

$$(A, B, C, D_0) = \big(f(H)_0, f(H)_1, f(Q)_0, f(Q)_1\big), \qquad T = \mathrm{Re}(k_H w).$$

These are the same functions as in the preceding construction: $k_H = y(H)/x(H)$, $w = h_S\tau$, and $\tau^2 = f(Q)$. The existing inverse of $f(Q)_0$ still excludes $S = P$; the tangent case $S = -P$ and all chart sign choices are retained.

Keep the three existing Laurent steps and scaled proxy,

$$rq_3 = q_1^2, \quad rq_7 = q_3^2, \quad rq_{15} = q_7^2, \quad q_1 = r^{-1}, \quad \kappa = q_{15} = r^{-15}, \quad ry = Tq_{15}, \quad y = T/r^{16}.$$

Replace the scalar tree by

$$n = B^2/r^2 + C^2/r, \qquad L = A^2/r + n^2/r^2,$$
$$J = D_0^2 + \kappa y^2, \qquad N = L^2 + \kappa J^2, \qquad W = 1 + N.$$

The physical pole input remains $b = 42rW^2$ and is passed through the same normalized scalar pole equations. Since every coefficient in these real norm expressions is positive, $N \geq 0$ and $W \geq 1$.

The two left nodes use exactly four rational coordinates:

$$z = B/r, \qquad rn = zB + C^2, \qquad u = n/r, \qquad rL = A^2 + un.$$

Each equation has degree at most two. The J and N nodes each retain a stored square of their right argument, so their inverse-weight terms are also genuinely quadratic in stored coordinates. The final addition $W = 1 + N$ is affine and introduces no coordinate. Relative to the preceding implementation, the former node $n_0 = 1 + A^2/r$ disappears; the former two-coordinate realization of $L = n_0^2 + rn^2$ is replaced by the displayed two-coordinate realization of the new L.

All new denominators are r, which is already positive and has an explicit rational inverse. Thus the rational extensions defining the new nodes are unique, and no new branch or zero-denominator domain is introduced. The shifted-point identity guard is not inferred from these scalar equations and is explicitly retained.

3.2 Odd contact valuations are already positive

Let p be a consumed prime outside the retained exceptional set $\mathcal{S} = \{2, 3, 5, 7, 13, 17\}$. The three-contact argument supplies an odd valuation $R = v_p(r)$. Its positivity need not be enforced by an all-left constant path in a norm tree.

Indeed, if $v_p(X) < 0$, the leading coefficient $11016 = 2^3 3^4 17$ is a p-unit. The three terms in the expanded polynomial for β then have distinct relevant smallest valuation, so

$$v_p(\beta) = 2v_p(X) < 0.$$

Since 32 is also a p-unit,

$$v_p(\beta - 1) = v_p(32\beta + 343) = 2v_p(X), \qquad v_p(r) = 6v_p(X),$$

which is even. If instead $v_p(X) \geq 0$, all three contact factors are integral, so $v_p(r) \geq 0$. Consequently oddness of R implies $R > 0$ for every consumed prime. This is a statement about the actual input polynomial; it would not follow for an arbitrary independently chosen rational β.

3.3 Soundness with an outer constant

At such a prime, 42 is a unit and the unchanged scalar pole implication gives

$$R + 2v_p(W) > 0, \qquad v_p(W) > -R/2.$$

Because $R > 0$, also $v_p(1) = 0 > -R/2$. Applying the ultrametric inequality to $N = W - 1$ therefore gives

$$v_p(N) > -R/2.$$

This step allows cancellation between 1 and N; no assertion of noncancellation at the final affine addition is made.

Within every binary norm node, the two rational summands have opposite valuation parity. In n their coefficient valuations are $-2R$ and $-R$; in L they are $-R$ and $-2R$; in both J and N they are 0 and $-15R$. Thus each node's valuation is exactly the smaller summand valuation. The argument also covers zero arguments by taking $v_p(0) = +\infty$.

Descending from N gives

$$v_p(L) > -R/4, \qquad v_p(J) > 29R/4.$$

The two left nodes give

$$v_p(A) > 3R/8, \qquad v_p(n) > 7R/8, \qquad v_p(B) > 23R/16, \qquad v_p(C) > 15R/16.$$

The right node gives the same bounds as before,

$$v_p(D_0) > 29R/8, \qquad v_p(y) > 89R/8, \qquad v_p(T) > 217R/8.$$

In particular all four coefficients of $f(H)$ and $f(Q)$ have positive integer valuation. Both points therefore belong to the formal group at every place above p, before the scalar proxy lemma is applied.

The retained two-place proxy lemma, including its nonunit fixed-coordinate case at 41, now gives

$$v_p(T) = v_p((a-1)/h).$$

Since $R \geq 1$ and rational valuations are integral, the strict bound $v_p(T) > 217R/8$ implies $v_p(T) \geq 28$. Since h is a nonzero integer,

$$v_p(a-1) \geq 28, \qquad M_{\mathcal{S}}(\beta)^{28} \mid a - 1.$$

This is the same uniform scalar order as before. No stronger bound $v_p(T) \geq 28R$ is asserted. In particular, the same scalar height consumer remains valid without changing the logarithm precision or introducing an exceptional prime.

3.4 Completeness and the order of choices

Fix an integral input, so r is a fixed positive integer. The inherited preparation chooses H first and then chooses Q sufficiently deeper at a finite set of primes. For fixed H, the retained function $T = \mathrm{Re}(k_H w)$ tends to zero as $Q \to O$, because $S \to P$, h_S stays bounded, and $\tau \to 0$. Arbitrarily large finite lower bounds on the four scalar coefficients and on T are therefore available in the same order as before.

For $p \mid r$, put $R = v_p(r) \geq 1$. The following explicit sufficient bounds establish the new local preparation:

$$v_p(A) \geq R + 1, \quad v_p(B), v_p(C) \geq 2R + 1, \quad v_p(D_0) \geq 4R + 1, \quad v_p(y) \geq 12R + 1.$$

They imply

$$v_p(n) \geq 2R + 2, \qquad v_p(L) \geq R + 2, \qquad v_p(J) \geq 8R + 2.$$

Consequently

$$v_p(N) \geq \min(2R+4, R+4) = R+4 > 0, \qquad W \equiv 1 \pmod{p}.$$

The lower bound for y is just the finite requirement $v_p(T) \geq 28R+1$, attainable after H is fixed. At a prime dividing 42 but not r, requiring the four coefficients and y to be divisible by p again gives $N \equiv 0$ and $W \equiv 1$ modulo p. These preparation statements use sufficiently deep formal neighborhoods; they do not assume good reduction at the small or ramified primes.

Thus $42rW^2$ is positive and nonzero, its reduced denominator is a square, and the same local squareclasses at $2, 3, 17$ and the same normalized pole realization are available. The inequality $W \geq 1$ retains the real preparation used previously. Once S is fixed, every changed node is a rational function of H with nonzero denominators at the prepared point. The finite lower bounds persist on an appropriate congruence class of h. Roots and scales are therefore still fixed before the contact-prime selection. No simultaneous prime selection or new global theorem is used.

3.5 Actual measured cost and finite checks

With the sixth-round detector and anchor parameters unchanged, the complete isolated pipeline gives

raw 409/377, generic compiler 399/367, $\boxed{\text{final 395 coordinates and 363 residuals}}$.

The pairs list coordinates and residuals. The independent gate replay checks all 348 source gates and the compiler operations; four further certified eliminations give the displayed final count. Every residual is quadratic, so the sum of their squares remains quartic. These are counts from the complete implementation, not estimates from the displayed abbreviations.

The standalone geometry checker reconstructs the actual changed scalar rows and their unchanged normalized pole consumer, as well as all retained point, difference, logarithm, scalar proxy, and nonzero-domain rows. It checks the exact input polynomial, affine valuation bounds, and finite completeness inequalities. Its 223 source assertions and the retained exact identity, tangent, and split-41 checks supplement this uniform proof; they do not establish the inherited global arithmetic.

4 The selected smaller anchor and unchanged arithmetic consumers

The starting construction is the frozen sixth-round package, with builder SHA256

`3f63cb782d52857446cf39f6e545aeb74f380d1347826eafa888cbb487738d51`.

The selected integrated source has builder SHA256

`78b54bcde24ff8a91091e102a4734c49878699f071d9230e5ce7a70ffd2114d0`.

The upstream source remains pinned at `036cebe4f1a4aa7ca89820a9b48f3629b4c364d6`. All inherited curve, rank, quaternion, height, pole, simultaneous-prime and Lucas inputs retain their previous qualifications. The new scalar and detector implementations have separate local proofs and source checks; this section checks their actual arithmetic consumers and the smaller anchor constant. Computational checks do not establish the inherited global inputs.

4.1 The exact retained anchor domain

Put $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, and $N(x+y\rho) = x^2 - 2y^2$. For the four root labels (n, a, t, b), write $C = \prod_i x(Q_i) = C_0 + C_1\rho$ and $B_0 = 4096C_0$. The scalar point is $S = [a]P$ and $Q = S - P = [a-1]P$. Its

unchanged chart and nonzero guard supply $a \neq 0$, $a-1 \neq 0$, $h_S \neq 0$, and an already stored coordinate $w = h_S\tau$ with $h_S^2 = x(S)$ and $\tau^2 = f(Q)$. Use

$$S_0 = 3^{2790}, \qquad \Theta = N(w) + 3S_0N(B_0 + X\rho), \qquad u^2 + 6v^2 - 2z^2 - 3\Theta^2 + 2S_0^2 = 0.$$

This is the same stored-root physical anchor as in round six, with its fixed exponent changed. Its exact characterization is

$$\text{all four root labels odd}, \qquad X \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}, \qquad 0 \neq a-1 \in 2 \cdot 3^{1394}\mathbb{Z}.$$

For clarity, the local order of this characterization is as follows. The actual quaternion criterion tests integrality at 2 and 3 of

$$\Theta/S_0 = N(w/3^{1395}) + 3N(B_0 + X\rho).$$

At 3 the norm is anisotropic modulo 3, so the two summands have opposite valuation parity; their separate integrality follows. The actual product leading-residue formula

$$v_3(C_0) = -\sum_i e(n_i), \qquad e(n) = 0 \ (n \text{ odd}), \quad e(n) = 2 + 2v_3(n) \ (n \text{ even})$$

then forces all labels odd before using any odd-product height identity. For odd a, h_S is a 3-adic unit and $v_3(w) = 1 + v_3(a-1)$; this gives the displayed divisibility. At 2, the integral odd formal series gives $v(z([j]P)) = 3/2 + v_2(j)$ for nonzero integers j. Thus the already proved oddness gives $v(h_S) = -3/2$ and $v(w) = v_2(a-1) \geq 1$, making both rational coefficients of w 2-integral. Also B_0 is 2-integral. The only remaining possibly nonintegral term is $-6X^2$, whose valuation is odd, so X is 2-integral. The converse follows by the same valuations and the quaternion criterion, including its zero-argument case. Sign choices of h_S and τ are harmless, and no square-root branch is added.

The actual source passages are `manuscripts/quartic_v10.tex`, lines 8132–8244 and 10973–11024; the prior packed-anchor leading-residue proof; and the sixth-round stored-root anchor proof. This specialization adds no new global assertion. Completeness chooses the positive scalar index $a-1$ arbitrarily divisible, so $3^{1394} \mid a-1$ is a finite preparation requirement compatible with the same later choices.

4.2 The three height consumers retain their hypotheses

The retained exceptional set is $\{2, 3, 5, 7, 13, 17\}$ and its product is 46410. Its inherited height constant remains exactly

$$H = 2^{110}(2^{32}3^9)^6\, 46410^{19} + 128.$$

Its bit length is 683; no decimal or logarithmic approximation is used in the accompanying numerical certificate.

The new scalar tree still gives $M_{\mathcal{S}}(\beta)^{28} \mid a-1$, so

$$h(\beta) \leq H|a-1|^{19/28}, \qquad \beta = 2401 + 2754(2X+1)^2.$$

The detector normalization keeps the actual orders $1, 12, 45$ at every consumed contact: its guard $\delta = 98t - 1$ is a unit there, so absorbing one factor of δ into each semantic test does not change the valuation conclusion. The precise arbitrary-order implication and its completeness are proved in the detector section; they are not inferred from a circuit count.

The long logarithm is still G_{12}. At every consumed prime $p \geq 11$ and formal depth $r \geq 1$, its omitted index $j \geq 13$ satisfies

$$4jr - (4j+1)/(p-1) \geq 45r,$$

using $(4j+1)/10 \le 4j-45$. The valuations tend to infinity, so the entire omitted tail has the required precision. The input logarithm is still G_2: its first omitted coefficient is $-14/195$, integral outside the retained exceptional set, and gives depth at least $12r$. For $j \ge 4$ the same factorial bound gives depth at least $12r$ because $(4j+1)/10 \le 4j-12$. Thus no new exceptional prime is needed. The primary inverse-root test of order 45 gives primary depth at least 45 and primary relative logarithm error at least 180. The existing weak-unit and two-anchor comparisons consequently give Lucas-label congruences of order 45 and input-label congruence of order 12, with their identity-input branch unchanged.

The three scales $1, C_0^2, (C_0+C_1)^2$ remain fixed positive rational squares after the four roots are chosen. The preceding product-height bound therefore remains

$$B^2 - 6 < 7H|N|^{19/k}, \qquad B = \max(|n|,|a|,|t|,|b|),$$

whenever the three contact radicals divide a nonzero integer N to order k. The actual odd-product and height consumers are `docs/quartic-523-540/addition.tex`, lines 286–350 and 421–477, with the already reviewed changed coordinate and scale constants. An optional projective-height improvement and an odd-exponent anchor lemma are recorded separately; neither is needed here.

In particular the literal Lucas polynomial is

$$F = t^2 - a(a-2)tb + (a-2)^2b^2 - 1.$$

For $B > 4$, its source bound is $|F| < 5B^4$. If $F \neq 0$, the order-forty-five comparison and the displayed height inequality give

$$B^{14/45} < 7H\,5^{19/45} + 6 < 24H.$$

The final estimate uses $5^{19/45} < 3$ and $H \ge 4$. If instead $F = 0$, the nonidentity root Z has $b \neq 0$, and the inherited Lucas theorem gives $\log B > |a| - 3$. Finally, if $n \neq \beta$, the nonzero reduced numerator of $n - \beta$ has absolute value at most $2e^{h(\beta)}B$. The order-twelve comparison and height inequality give the input estimate used below.

4.3 Exact margins at exponent 2790

The selected construction uses the particularly simple even exponent $E = 2790$, scalar order 28, input order 12 and high order 45. This choice already works with the preceding $7H$ product-height bound and its $24H$ Lucas consequence. Thus the optional projective-height improvement in the separate research note is not a new dependency of the selected construction. The odd-exponent generalization is also not needed for this selection: the physical coefficient remains $\gamma = 3$.

Write $c = |a-1|$. Here $q = 1395$, $c_* = 2 \cdot 3^{1394}$. Three small exact-integer checks are

$$3^{53} > 2^{84}, \qquad H < 2^{683}, \qquad 3H < 2^{684}.$$

They give the transparent strict lower bounds

$$\log_2 c > \frac{117149}{53}, \qquad \log_2 |a| > \frac{117096}{53}.$$

Indeed $c \ge c_*$ and $|a| \ge c - 1 \ge c/2$. Consequently

$$\frac{9}{28}\frac{117149}{53} > 690 > \log_2(80H), \qquad \frac{14}{45}\frac{117096}{53} > 687 > \log_2(24H).$$

The first inequality proves $h(\beta) < c/80 \le |a|/40$; the second contradicts the unchanged consequence $B^{14/45} < 24H$ of $F \neq 0$. Also $\log(40H) < 689$ and $40 \cdot 689 < 2^{15} < |a|$, giving the separate logarithmic margin. The input-order-twelve contradiction is exactly

$$\log B < \frac{12}{5}\log(40H) + \frac{19}{5}h(\beta) < \frac{31}{200}|a| < |a| - 3.$$

The last comparison holds already when $|a| > 4$.

All inequalities are strict. The companion checker also verifies the stronger direct integer tests $c_*^9 > (80H)^{28}$ and $(c_* - 1)^{14} > (24H)^{45}$ without logarithmic approximation. Thus there is no rounding assumption about the 683-bit constant. The fixed parameter can be written $S_0 = c_0^6$ with $c_0 = 3^{465}$; any subsequent nonzero diagonal coordinate rescaling using c_0 requires only its separate exact algebraic equivalence audit.

The two contradictions therefore force $n = \beta \in \mathbb{Z}$. Together with $X \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$, this gives $X \in \mathbb{Z}$: at every other prime the leading coefficient of $\beta = 11016X^2 + 11016X + 5155$ has valuation zero, except valuation one at 17, and a negative valuation of X would in either case force a negative valuation of β. Completeness uses the same integral input map, odd root labels, finite scalar preparation, primary progression $\Pi^{44} \mid h$, positive square scales, Gaussian positivity witnesses, and subsequent contact-prime selection. The selected smaller anchor changes only the finite divisibility already checked above. No independent exponent-ten anchor requirement is silently discarded: its actual Lucas consumer has been checked directly at the retained high order forty-five.

5 Balancing coefficients without adding variables

The preceding arguments allow the fixed anchor integer

$$S_0 = 3^{2790} = c^6, \qquad c = 3^{465}.$$

The core produced by the selected source has 392 active rational coordinates and 360 quadratic residuals. Its integer coefficients are already explicit after specializing S_0. A diagonal rational change of twelve existing coordinates makes those coefficients substantially smaller without introducing a constant-chain witness.

5.1 The exact change of coordinates

Let x_i denote a coordinate of the compiled source system and y_i its replacement. The distinguished input is unchanged: $x_0 = y_0 = X$. All unlisted coordinates are unchanged. For the listed coordinates set $x_i = c^{e_i} y_i$ with the following exponents:

Source coordinates	Number	e_i
The two coefficients of h_S	2	1
The two stored coefficients of $x(S)$	2	1
The two coefficients of $w = h_S \tau$	2	2
The two coefficients of $f(S - P)$	2	1
The three translated anchor witnesses U, V, W	3	4
The materialized physical anchor Θ	1	4

The coordinate names and integer source identifiers are bound in the machine-readable certificate. The coefficients of h_S and $x(S)$ are intentionally scaled by the same factor, although the old identity is $x(S) = h_S^2$. Their new defining equations incorporate that factor; the new coordinates are decoded before being interpreted as the old point functions. No semantic identity is silently assumed unchanged.

Since c is a fixed nonzero rational integer, this diagonal map is a bijection on $\mathbb{Q}^{392}$. Its inverse divides a listed coordinate by c^{e_i}. It fixes the input and does not introduce any square-root choice, new quantified parameter, or arithmetic hypothesis.

Let $R_j(x, S_0)$ be an integer quadratic residual of the compiled source. Substitute $x_i = c^{e_i} y_i$ and $S_0 = c^6$, combine like monomials, and let $d_j \neq 0$ be the signed integer content of the resulting row, with the sign chosen to make the first coefficient positive. Set

$$Q_j(y) = \frac{R_j((c^{e_i} y_i)_i, c^6)}{d_j}.$$

Every Q_j is a primitive integer polynomial of degree at most two. The delivered polynomial is exactly

$$P_{392}(y) = \sum_{j=1}^{360} Q_j(y)^2.$$

Over $\mathbb{Q}$, a sum of squares vanishes if and only if each summand vanishes. Thus

$$P_{392}(y) = 0 \quad \Longleftrightarrow \quad R_j((c^{e_i} y_i)_i, c^6) = 0 \quad \text{for every } j.$$

Combining this equivalence with the diagonal bijection proves equality of the existential projections to X. The new polynomial need not be a single constant multiple of the old sum of squares: its individual rows have been rescaled, which preserves their common zero set.

5.2 Why the exponent drops

Before the change, the fixed parameter occurs with exponent six in c. After substitution, each affected row has a common power of c whose removal leaves a span at most two in its remaining c-exponents. For example, the physical anchor row contains Θ and $N(w)$ at exponent four and $S_0 N(B_0 + X\sqrt{2})$ at exponent six. Dividing by its common factor leaves exponent two. In the translated anchor row, the quadratic witness terms have exponent eight and the $S_0 U$ term has exponent ten, again leaving a span of two.

The shifted-point equations, the root chart equations, and the scalar leaves have the same bounded span. All of these statements are checked on the actual compiled rows. Consequently their squares involve at most c^4 in the large coefficient, in place of the former $S_0^2 = c^{12}$.

For the selected source, exact full expansion gives

$$\max |\operatorname{coeff}(P_{392})| = 1 + 40532396646334464\, c^4 = 1 + 40532396646334464\, 3^{1860}.$$

This integer has 905 decimal digits and 3004 bits. The complete quartic has 26,301 nonzero monomials, content one, and coefficient one on X^4. Every coordinate occurs. The two constant-chain coordinates used in the preceding coefficient-controlled construction are therefore unnecessary at its former 978-digit cap.

The diagonal pattern was suggested by a linear optimization of coefficient valuations. That floating-point search is not a verification dependency. The selected pattern uses only the displayed integer exponents; the delivered producer uses Python's standard library and exact integers. A separately authored checker reconstructs all 360 substituted rows, verifies each nonzero row divisor and the input-fixed diagonal map, independently expands the complete sum of squares, and compares every coefficient of the text table. The search makes no optimality claim.

6 Verification and coding-agent handoff

The independent source checks cover 384 detector assertions, seven Laurent identities, three normalization identities and two mixed-product identities; 223 geometry assertions; separate scalar reconstructions

with 70 and 69 source assertions; and 58 arithmetic source/finite checks. The uniform arguments were reviewed separately from those finite checks. The scalar reviews check formal membership before use of the two-place proxy, including the retained case at 41. The detector review checks the actual common-place conclusion, unchanged R, off-contact parity, and the real and small-prime completeness limits.

The independent diagonal audit reconstructs all 360 changed rows and their nonzero divisors, the input-fixed coordinate bijection, the full 26,301-monomial sum of squares, coordinate compaction, primitive content, the exact largest coefficient, and every coefficient in the text table. The unbalanced core and three optional lifts have their own complete sum-of-squares and text-table audits. The parameter-reader checks exercise actual arithmetic at several exponents and reject inconsistent declarations; a changed metadata label alone is insufficient.

The PR payload contains source scripts, proofs, the selected decimal table, compact evidence, and deterministic expected-output hashes. Large intermediate JSON files are regenerated in a fresh output directory. The reproduction wrapper copies only source files into that directory, runs the full construction and audits, and compares every recorded deterministic artifact. It leaves the tracked proposal files unchanged. Its computation uses Python 3.11 or newer and Node.js, with their standard libraries only and no network dependency. A TeX installation is needed only to rebuild this PDF.

The handoff provides an additive patch, a repository overlay, a draft PR title and body, exact upstream dependency hashes, and commands for validation. A coding agent should compare the pinned baseline with its current checkout, read the repository's actual `AGENTS.md` and audit protocol, and reconcile any intervening changes before proposing integration. The historical Version 10 review register must remain historical until any new manuscript application has been reviewed.

The optional parity and projective-height note is included as research material. Its improved constants are not dependencies of the selected 392-variable construction. The lower bounds and negative search results are explicitly limited to the stated circuit search classes.